

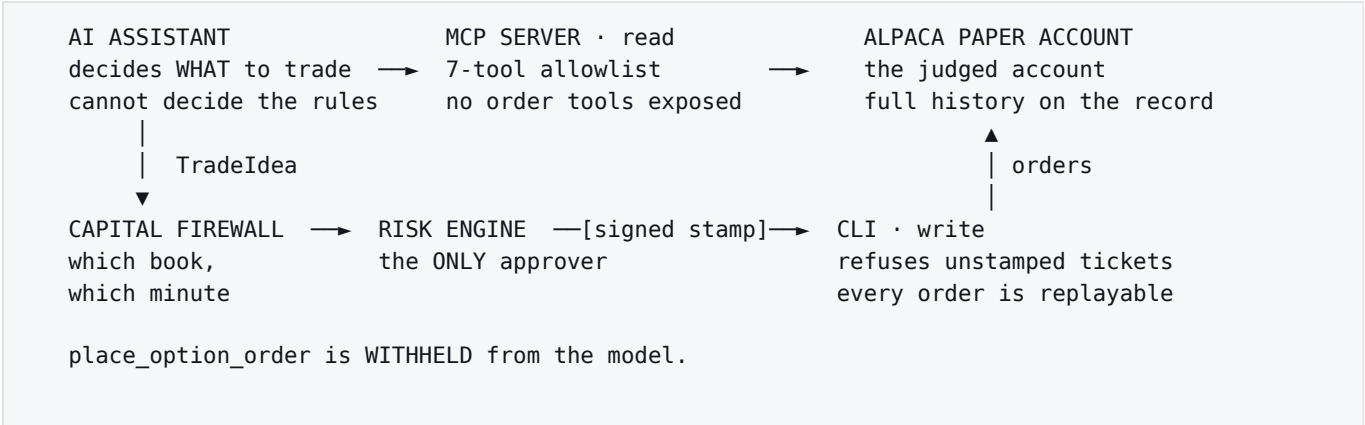
Three books, one account.

A resident short-premium book, a transient day book, and an overnight events book bounded by size rather than by the firewall. This is the whole system, from the outer frame down to each gate — and every threshold in it was moved at least once, always by measurement.

REPO	BOOKS	JUDGED ACCOUNT	WINDOW
alpaca-hackathon	carry · intraday · events	PA3TSH9YTFJL	28 Aug – 4 Sep 2026

One account is the only shared resource

Everything else in the system exists to decide who may touch it, with how much, and at which minute. Alpaca is reached three ways, deliberately, for three different jobs.



Reads and writes take different paths on purpose. The agent queries Alpaca's MCP server directly and reasons over live state — that is real autonomy, not a scripted pipeline with a language model bolted on. But the server's own order-placing tools are never exposed to it. An agent that can place a raw order can bypass the firewall, and the whole point of the firewall is that nothing can.

SURFACE	JOB	WHY THIS ONE
MCP server	read	account, positions, orders, clock, quotes, news — seven tools, allowlisted
Alpaca CLI	write, and the backtest data path	every order is a shell command you can paste into a terminal and replay
Trading API (REST)	the judged profile's broker	broker.primary: rest, and the only path with websocket streaming

One resident, one transient tenant, one overnight guest

Carry and intraday are uncorrelated in signal and opposite in vol exposure. That is the real portfolio argument for running both — and the reason the boundary between them has to be enforced rather than assumed.

RESIDENT · FIREWALL TENANT vol_carry Sells the IV–RV spread as defined-risk structures and holds them. Theta accrues on calendar days, weekend included; a nightly round trip would pay the spread for nothing. EDGE variance risk premium SHAPE iron condor, 0.25Δ shorts HOLD 3–10 sessions, 7–14 DTE WINDOW 10:00 – 15:00 ET UNIVERSE 14 index & sector ETFs RISK short gamma into a move	TRANSIENT · FIREWALL TENANT intraday_momentum Momentum expressed through options. The option is leverage and defined risk — it is not the source of edge. Flat by 15:10, five minutes ahead of the cutoff, every session. EDGE directional continuation SHAPE long option · debit vertical HOLD minutes, 0–2 DTE WINDOW 09:45 – 14:45 ET UNIVERSE 8 index ETFs RISK spread cost, signal decay	OVERNIGHT · BOUNDED BY SIZE earnings_event_directional The only book whose gate opens on a <i>date</i>. It leases no capital from the firewall — a book whose entire life is one overnight hold does not fit the tenancy model, so three size limits bound it instead. EDGE implied vs realised reaction SHAPE condor if rich, vertical if cheap HOLD one night WINDOW arm 15:50 · flat 09:45 ET BOUNDS 1.2% / name · 4% / night · 10 contracts RISK the gap — bounded by structure
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A FOURTH BOOK, ENABLED AND UNABLE TO TRADE

event_premium (book opportunistic) is enabled: true in config and cannot open a position: _transient_scan always acquires the capital lease as INTRADAY, so the opportunistic book's may_open check always fails. It is documented as enabled because that is what the config says, not because it trades.

Forty-six cycles, and one of them is a hard stop

Every row below is a firewall boundary. The runner re-fires any cycle whose time has passed, so a crash at 15:10 cannot skip the 15:15 cutoff.

04:00–16:00	events_watch	hourly, 13 polls. Read the wire on names reporting this week
09:15	discover	attention sweep + macro regime read, pre-market
09:30	– bell –	OPEN_SETTLE: nothing opens, quotes are wide
09:45	events_flatten	close last night's spreads into the IV crush, before the day books compete for the same buying power
10:00–11:15	intraday_scan	every 15 minutes
10:00	carry_scan	resident book reserves capital, hunts rich premium
11:30–15:10	manage_positions	11 passes: profit target, stop, DTE floor, defensive close
13:30–14:45	intraday_scan	every 15 minutes (11:30–13:30 skipped by the strategy's own time gate)
13:50, 14:40	carry_scan	two further passes
15:15	intraday_cutoff	HARD: cancel → liquidate TRANSIENT ONLY → poll until confirmed flat
15:45	carry_verify	zero transient exposure, fresh Reg T, carry margin covered
15:50	events_arm	open tonight's earnings spreads, one night only
16:00	– bell –	the carry book is HELD. There is no nightly exit.
16:10 / 16:20	report	performance report, then the LLM-written daily report and its evaluator

Twelve intraday scans, not two. A VWAP cross is an event that lasts minutes: two scans a day observed roughly 30 of 390 session minutes and required the cross to land inside one of them. These times mirror `backtest.session_times_et` exactly — when they diverged, the live agent saw a sixth of what the backtest was tuned against.

Two dated controls fire from the clock rather than from the daily schedule, and the second is checked on every runner poll:

2026-09-02 20:00 UTC	vol_carry entry cutoff	stop opening CARRY structures only
2026-09-04 13:45 UTC	submission_flatten	close EVERYTHING, then refuse every book

WHAT THAT SECOND DATE MEANS IN PRACTICE

13:45 UTC is **09:45 ET on 4 September**, and the intraday entry window opens at 09:45. **4 September yields zero new trades.** The tradeable sessions are 2 and 3 September.

The reservation comes off the top

Alpaca grants 4× day-trading buying power but only 2× Reg T overnight. Intraday leverage still on the books at 16:00 is a broker-forced liquidation — unrecoverable inside a one-week judged window. Two layers make the overlap impossible.

Layer 1 — temporal

firewall/clock.py derives every boundary from one ET phase machine:

```
CLOSED → OPEN_SETTLE → INTRADAY → ACTIVE → CARRY_ONLY → WIND_DOWN
      → INTRADAY_CUTOFF → CARRY_VERIFY → CLOSED
```

Phase.intraday_may_open and Phase.carry_may_open are the only two questions anything asks it. Boundaries must be strictly increasing and the cutoff must sit at least 15 minutes before verification, both validated on load.

Layer 2 — capital

```
Reg T buying power  (fresh poll, never a cached number)
|
|— carry requirement —————→ reserved FIRST, never lent out
|   live marks on the legs the ledger attributes to `carry`, × 1.25 cushion,
|   capped at carry_max_equity_pct 0.50
|
|— what is left × transient_utilisation 0.50
    ^ transient_max_equity_pct 0.15  → transient lease
```

The property, asserted directly in tests/test_firewall.py: for every level of carry usage, $\text{carry_claim} + \text{transient_claim} \leq \text{regt_buying_power}$, and the transient lease shrinks monotonically as the resident book grows. The transient lease is what is *left*, not a share of the whole.

The position ledger

Once the carry book became resident, "flat" stopped being a property of the account and became a property of a *book*. firewall/ledger.py persists a leg → book map to disk, so a restart at 15:10 cannot cause the 15:15 cutoff to liquidate a multi-session iron condor.

THE CONSERVATIVE DEFAULT

Unattributed legs are treated as transient. A leg the ledger has never seen is by definition not something the system deliberately chose to hold overnight. Closing it at 15:15 is the recoverable error; carrying it into the close is the one that is not.

What the cutoff actually guarantees

- **Liquidation is confirmed, not requested.** A 200 from close_all_positions means accepted, not filled — so it polls, four attempts, five seconds apart.
- **Working orders count as "not flat".** A resting order that fills at 15:59 is unexpected exposure into the close.
- **Residual transient positions at 15:45 disable the transient books for the following session.** This used to trigger on any open order account-wide — so a single resting carry entry at the sign-off cost the next day's intraday book an entire session. A working order is now cancelled by the emergency cutoff and noted, without the ratchet.

Four hard gates, not a score

A blended score would let a rich-IV reading paper over an earnings date. That is precisely the trade a short-premium book must never take, so every gate is hard and a candidate that passes three of four does not trade.

01 · PREMIUM

Is vol rich?

IV rank says vol is high for this name. The IV–RV spread says you are being paid more than the stock has actually been moving. **The spread is the edge; the rank is only a regime filter.** Both required — either alone is a known false-positive generator.

$$\text{iv_rank} \geq 0.35 \text{ AND } (\text{iv} - \text{rv}) \geq 0.03$$

02 · TREND

Is it going nowhere?

Short premium is short movement. A name that is going somewhere is disqualified before anything else is measured — and this is the same measurement that fires the momentum strategy, which is what makes the two mutually exclusive by construction. A test asserts they never co-fire.

$$\text{adx} \leq 25 \text{ AND } |\text{trend}| \leq 0.60$$

03 · EVENT

Hard exclusions

Earnings inside the expiry window, or an ex-dividend date under a short call. No model input, no scoring. IV is elevated *because* of the event, so that premium is fair — selling it is selling the one thing the market has priced correctly.

$$\text{no earnings} \leq \text{dte_max} \cdot \text{no ex-div with a short call}$$

04 · MACRO LENS

Shared, or idiosyncratic?

The question a z-score cannot answer, and the one place a language model genuinely earns its seat. Sector-wide IV elevation with no name-specific catalyst is *shared* — exactly the premium the thesis targets. A name repricing on its own news is *idiosyncratic*: the distribution has a fat tail the model cannot see. Veto.

$$\text{guidance} \neq \text{stand_down} \text{ AND } \text{symbol not flagged}$$

Then economics, and it is not a nicety: a four-leg condor crosses **eight half-spreads** on a round trip.

$$\text{round-trip spread} \leq 20\% \text{ of the credit received} \cdot \text{relative spread} \leq 10\%$$

THE IV-RANK FLOOR WAS 0.70, AND IT REJECTED 304 OF 304 CANDIDATES

Every threshold in this document was moved at least once, and always by measurement. A gate that rejects everything is not conservative — it is a book that does not exist. The rejection log is what made that visible: `oaa backtest --why 15` groups by reason and normalises the numbers inside each one, so "IV rank 19% below the floor" and "IV rank 4% below the floor" count as one finding rather than two.

The exit pair sets the hit rate you need

Breakeven hit rate = stop ÷ (target + stop). It is arithmetic, and it is decided before any trade is taken.

WAS · 30% TARGET / 2.0× STOP

87.0%

needed before any spread. The observed win rate was 88% — one point of margin. Worse, the 20% cost ceiling implied 95.7%, so the gate was admitting trades the arithmetic said would lose.

NOW · 50% TARGET / 1.5× STOP

75.0%

The same 20% cost ceiling now implies 85% — three points of margin. Same gate, opposite meaning, because the exits changed underneath it.

EXIT TRIGGER	ACTION
\$450 hard dollar stop	close — checked first
50% of max profit	close
loss reaches 1.5× credit	close
DTE floor (3)	close regardless of P&L — gamma rises sharply into expiry
underlying touches a short strike	close the tested side
macro lens flags the name mid-hold	close at the next open
submission flatten	close everything, unconditionally

A STOP SET ABOVE THE LOSS DISTRIBUTION IS DECORATION

The dollar stop was **\$900 and never fired**. The six largest losses were \$780–868 — all sitting just underneath it — and every one exited on the DTE floor instead, which realises whatever has accumulated rather than capping it. \$450 is a stop that can be reached.

Why 7–14 DTE, and not the conventional 20–45

Theta is steeply non-linear into expiry. At 30 DTE a five-session hold captures roughly 20–25% of the structure's remaining decay; at 10 DTE it captures the majority. The judged window is the constraint, and the structure's life has to fit inside it.

The cost is gamma, and it is real: short-dated short-premium positions reprice violently when the underlying approaches a short strike. Three controls offset it partially, not completely — 0.25Δ short strikes, the position-count cap, and the defensive exit on a short-strike touch.

SAY THIS OUT LOUD

This is a deliberate trade of tail risk for realised P&L inside the window. It is the right call for a seven-day judged period and the wrong call for a live book. A judge who sees 10-DTE condors will either think you do not know the convention or that you knowingly departed from it — and the second impression is only available if you write it down.

Eight vetoes in a row is not a strategy

Each gate passed roughly 70% of candidates. Stacked as hard vetoes that is $0.7^8 \approx 6\%$ — a book arithmetically designed almost never to fire. Measured over 864 candidates across 9 sessions: 424 died on the volume gate alone, and **zero** survived the whole chain. Loosening any one gate simply promoted the next one to being the wall.

The fix: two hard gates, everything else votes

HARD VWAP trigger It decides <i>direction</i>; without it there is no trade to make. A session VWAP cross, or price on the band within $2.0\times$ the session's usual dispersion, searched over the last 15 one-minute bars — exactly one polling gap, so a cross cannot fall between two cycles unseen.	HARD Spread gate It is economics, not evidence. $(ask-bid)/mid \leq 4\%$ and round-trip spread $\leq 30\%$ of target. The ceiling was 2% until measurement showed the tightest quote in the universe was 2.59% — the gate was rejecting the market rather than selecting within it.	THREE OF SEVEN The voters Volume $z \geq 0.5$ in the same time-of-day bucket · 2-bar persistence · Bollinger <i>width</i> rising · higher-timeframe agreement · term-structure slope in band · RSI not at 80/20 · catalyst score. needed = $\min(3, \text{possible})$.
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"UNMEASURABLE" IS NOT "FAILED"

With no time-of-day volume baseline yet, volume simply does not vote rather than blocking the trade. Demoting the catalyst to a vote exposed a second bug worth recording: the catalyst gate returned `ok()` immediately when `required=False`, before measuring anything — so it always voted yes and the fifth confirmation was free. It now always measures, and only `required` decides whether a failure is a veto or a lost vote.

Three indicators, three different questions

Only VWAP has an opinion on direction. RSI and Bollinger *position* are mean-reversion measures and would contradict a VWAP momentum trigger on every signal — price breaking above VWAP is simultaneously "momentum, buy" and "overbought, fade". Width, not position: band width is a volatility-regime read and is direction-agnostic. RSI is one-sided at 80/20 rather than 70/30, because strong momentum carries elevated RSI for long stretches and a tight threshold rejects exactly the trades this book exists to take.

The constraint that shapes everything: spread cost

QUOTE WIDTH	ROUND TRIP / CONTRACT	SHARE OF A \$20 TARGET
\$0.01 SPY ATM	\$2	10%
\$0.02	\$4	20%
\$0.05	\$10	50%
\$0.10 typical single name	\$20	100% — no edge survives

Index products only. Not a preference — arithmetic. Eight ETFs: SPY, QQQ, IWM, DIA, XLF, XLE, TLT, GLD.

Selection, and the exits

EXPECTED MOVE	IMPLIED VOL	SELECTION	REASONING
small	low	ATM, 0–1 DTE	max gamma, cheap
small	elevated	ATM, longer dated	less premium burn if the move stalls
large	low	slightly OTM	better convexity per dollar
large	elevated	debit vertical	caps the cost of the elevated IV; sells the expensive wing
any	$iv_rank \geq 0.85$	NO TRADE	paying for a move already priced in

Exits: 10% target, 15% stop, 20-minute time stop, flat by 15:10. The stop is *wider* than the target deliberately — option premium is noisy and a tight stop is hit by spread flicker alone — which implies a breakeven hit rate near 60%, computed and displayed against the actual.

KNOWN, AND STILL LIVE

`exit_on_vwap_recross` is still true in `config/strategies/intraday_momentum.yaml`. The A/B measured that rule at **0 wins in 55 attempts**.

The only book whose gate opens on a date

The other two wait for a market condition to cross a threshold, and the recurring failure is that it never crosses. Broadcom reports on 2 September whether or not any indicator agrees. The gate opens on schedule; the pipeline only decides whether the trade is worth taking.

The model proposes. A file confirms.

Featherless serves open-weight models whose weights were frozen months ago. Asked to list next week's reporters, one answers fluently and sometimes wrongly — and **a wrong date is not a bad trade, it is a position opened against no event at all**. So the model only proposes candidates; `config/events/earnings_calendar.json` confirms every one of them, and anything unverified is logged and never armed.

STAGE	WHAT IT DOES
<code>calendar.py</code>	model proposes → the calendar file disposes. Unconfirmed names never arm
<code>volscreen.py</code>	prices the ATM straddle in the expiry <i>containing</i> the print; ranks by divergence against the last four actual reactions. Implied move $\geq 3\%$, relative spread $\leq 25\%$, top 10
<code>watch.py</code>	hourly from three days out. Alpaca news + StockTwits, deduplicated by hash — a quiet name makes no model call . Notes kept at salience ≥ 0.35 , aged out at 10 days, capped at 40
<code>direction.py</code>	one call per name. Direction, confidence, and the evidence it cited. Abstention is a valid and expected answer
<code>sizing.py</code>	confidence maps linearly onto size, then bounded per trade, per night, and by an absolute contract cap
<code>strategy.py</code>	builds the structure the divergence calls for, behind three interlocks

The correction that mattered: we measured volatility and traded direction

A directional structure bought at a fair-to-rich implied move, on a direction call no better than a coin flip, returns minus the round trip in expectation. That is arithmetic, not a sample-size problem. So the *expression follows the sign of the divergence*:

IMPLIED MOVE ÷ MEDIAN REALISED REACTION	STRUCTURE	WHERE THE EDGE IS
≥ 1.35 rich	iron condor, both shorts <i>outside</i> the implied move	the overpricing itself
$0.80 - 1.35$ fair	no trade	none measurable — and this band is where the book used to do all of its trading
≤ 0.80 cheap	debit vertical, in the called direction	the underpricing

The thresholds sit well clear of 1.0 because the ratio is computed off four prints — a name at 1.10 is inside its own sampling error. Strikes are placed by **distance, not delta** (shorts at spot $\pm 1.0 \times$ implied move): the front-weekly surface across a print is too deformed for delta to stand in for distance, and where the shorts sit relative to that move *is* the thesis. A coarse strike ladder that snaps one back inside is declined, not quietly downgraded.

The direction call is a tilt, not the thesis. It picks the side of the vertical, and on a condor it pushes the threatened short further out — never pulls the other one in, because collecting more premium by moving a short inside the implied move surrenders the one property the structure is built on.

Two roles, two models

watch triage	Qwen3-8B	~40 calls/day
direction call	Qwen3-32B	once per name

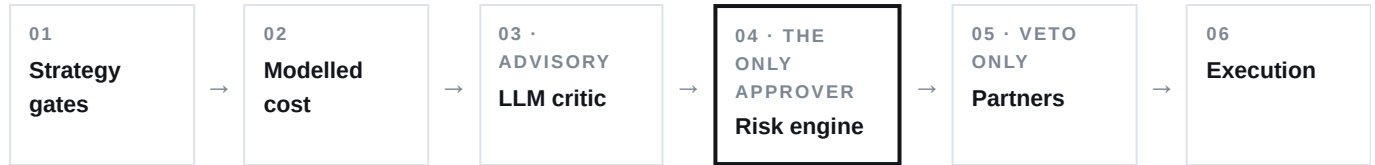
The triage is narrow, schema-bound and usually right to say "immaterial". Running the big model on it spends the direction call's budget on noise.

What bounds the model

- **Confidence floor 0.55** — and it sets the size, so a marginal call trades small.
- **Evidence required** — a confident call citing nothing is a guess with a number attached.
- **Abstention rate journalled** — zero abstentions logs a warning. A model that never declines is not filtering.
- **Evidence is untrusted text** — fenced, stripped, truncated, parsed to a fixed schema, every field clamped.

Only one component can say yes

The critic scores and writes reasoning. Partner adapters may veto. Neither can approve. Between the idea and the broker sits a deterministic engine, and execution refuses any ticket it has not stamped.



Stage 1 returns either a TradeIdea or a rejection carrying the gate that stopped it and the metric it measured. Stage 2 attaches a full modelled round trip — OCC clearing, ORF, FINRA CAT both sides, TAF and SEC on sells, exchange fees, margin interest — *before* anything judges it, so a rejection still carries what the trade would have cost. Stage 4 is fifteen deterministic checks in a fixed order:

```

firewall → halted → market_open → undefined_risk → unknown_risk → leg_count →
duplicate_legs → max_positions → max_new_per_day → duplicate_structure →
reentry_cooldown → concentration → sizing → portfolio_risk → cash
  
```

TWO OF THOSE EXIST BECAUSE BROKERS NET IDENTICAL OPTION SYMBOLS

Re-opening the same condor **doubles one position** rather than creating a second, so every count-based limit was blind to it. `duplicate_structure` refuses a structure whose every leg is already held on the same side; `reentry_cooldown` catches the near-duplicate once spot has moved enough to change the strikes.

- **Sizing is from max loss, not from capital.** A structure without a computable `max_loss` is refused outright (`allow_undefined_risk: false`) — which is what makes a stray undefined-risk idea impossible to execute by accident.
- **Portfolio Greek caps** are checked across the whole account, not per trade. A greek block of exact zeros is treated as *missing*, not as a measurement: the indicative feed serves those, and they turned three caps green on a book that had never been measured.
- **Atomic combos by default** — one spread crossed, not four — with a deterministic `client_order_id` and a pre-submit existence check, so a retry cannot double-fill.
- **On the judged profile the limit posts at the far touch.** At the 0.5 default with the chase disabled, every order rested half a spread from the touch with no second attempt, and simply never filled. The chase itself stays off: it cancels without checking the cancel succeeded, then resubmits under a new order id — a race that can double the risk-approved size.
- **Ledger registration is what makes 15:15 correct.** Every filled leg is attributed to its book the moment it opens.

Built from the first commit, not retrofitted for the deck

ARTEFACT	WHY IT EXISTS
Gate rejection log oaa gates	Which gate vetoed each candidate and every metric it measured — 72,753 rejections across the Feb–Aug replay against 441 trades taken. The highest-value judging artefact: it shows an agent declining trades, with the number that caused it.
Modelled cost line	Gross P&L, modelled cost, net — side by side, with the spread charged <i>inside</i> the fills rather than added beside them. Paper charges no fees and fills at mid.
Breakeven hit rate	Computed from the target/stop asymmetry and shown against the actual rate, rather than assumed away.
Equity curve	Snapshotted through the session, so the curve is dense enough to be the primary visual evidence.
Macro lens archive	Regime, flagged symbols and themes, per cycle — the reasoning trail behind every stand-down.
Daily report oaa daily-report	An LLM-written end-of-day report <i>and</i> an evaluator score for it. The only page that says what the system thinks is wrong with itself, and it is published unguarded on both dashboard builds.
Watch dossiers	Dated pre-print notes per name, published to the public Events tab. Retired dossiers are deliberately never published: post-print commentary beside names still being watched reads as a pre-print call.

Six things that are true and should not be hidden

A judge who spots an unacknowledged paper-fill advantage forms a much worse impression than one who reads that it was modelled.

The credit-to-width audit

A share of the carry book's replayed P&L sits in structures whose recorded credit exceeded their own width — arithmetically impossible for a condor, since a condor cannot collect more credit than it can lose. Reproduced independently by computing $\text{credit} \div \text{width}$ per trade, 47 of 198 carry trades carried the large majority of the book's P&L. The gated re-run is far smaller than the ungated headline, and `risk.max_credit_to_width` is **not implemented in the working tree** — so a run started from it reproduces the ungated number. Quote the gated figure, or state the provenance.

And five more

- **Exit code is not shared.** Entry runs the same objects live and in replay. Exit is reimplemented in the backtest engine, and nothing asserts the two agree.
- **Intraday replay marks are coarse** relative to a 20-minute hold, so that book's replay P&L reads for candidate flow, not for edge.
- **Paper fills flatter.** Mid fills, no queue position, no partials — and spread cost is the intraday book's primary loss mechanism, which is exactly what paper does not simulate.
- **Five sessions is not a sample.** Expect 10–20 trades in the judged window. That cannot distinguish edge from luck in either direction.
- **The events book has no usable backtest.** Four prints per name is not a sample you can walk forward, and the divergence ratio is a ranking device rather than an edge estimate.
- **Short-dated short premium is a deliberate tail-risk trade.** 7–14 DTE fits the judged window and would be the wrong call for a live book.

OPERATIONAL, AND IT OUTRANKS THE REST

There is no pidfile, lockfile or singleton guard. Two runners append to one journal, race one `position_ledger.json`, and hold separate in-memory caches — so one can serve pre-fix data hours after a fix lands, and every conclusion drawn from the journal while both are live is suspect. **Run exactly one instance.**